



Correlation and Covariance

Understanding how two variables move together

Simple explanations • worked examples • practical use cases



1. Covariance

Covariance shows whether two numerical variables generally move in the same direction or in opposite directions.

$$\text{Cov}(X,Y) = \text{average of } [(X - \text{mean } X)(Y - \text{mean } Y)]$$

- Positive covariance: both variables tend to rise or fall together.
- Negative covariance: one tends to rise when the other falls.
- Near-zero covariance: little linear co-movement is visible.

Its size depends on the measurement units, so it is difficult to compare across datasets.

2. Correlation

Correlation standardizes covariance, making the strength and direction easier to interpret.

$$r = \text{Cov}(X,Y) / (\sigma X \times \sigma Y), \text{ where } -1 \leq r \leq +1$$

Correlation	Meaning
+1	Perfect positive linear relationship
Around +0.8	Strong positive linear relationship
Around 0	Little or no linear relationship
Around -0.8	Strong negative linear relationship
-1	Perfect negative linear relationship

3. Worked banking examples

Variables	Correlation direction	Reason
Income and savings	Positive	Higher income may support higher savings
Interest rate and loan demand	Negative	Higher borrowing cost may reduce demand
Account number and balance	Near zero	Identifier has no meaningful numeric relationship
Transaction frequency and fee income	Positive	More eligible activity may increase fee income

A scatter plot should be inspected alongside correlation because outliers, clusters and curved patterns can produce misleading summaries.

4. What correlation does not mean

- Correlation does not prove that one variable causes the other.
- A third variable may influence both observed variables.
- A correlation near zero does not rule out a curved relationship.
- A strong correlation can be driven by a few extreme observations.
- Combining separate customer segments can hide or reverse relationships.

Use correlation as evidence of association. Establishing causation requires stronger design, domain knowledge and controls.